

A Field Project Report On
“Attendance Management System”

Submitted

In partial fulfilment of the requirements for the award of the degree

BACHELOR OF TECHNOLOGY

In

COMPUTER SCIENCE AND ENGINEERING

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**VIGNAN'S FOUNDATION FOR SCIENCE, TECHNOLOGY &
RESEARCH**

(Deemed to be University)

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CERTIFICATE

This is to certify that the Field Project entitled “**Attendance Management System**” that is being submitted by [231FA04843],[**A. Vamsi Krishna**], [231FA04857],[**K.Venateswarlu**],[231FA04863],[**D.Tapaswi**],[231FA04868],[**K.Sasank sai**] in partial fulfilment of the requirements for the degree of **Bachelor of Technology (B.Tech.) in Computer Science and Engineering** at the Department of Computer Science and Engineering, Vignan’s Foundation for Science, Technology and Research (Deemed to be University), Vadlamudi, Guntur District, Andhra Pradesh, India.

This is a bonafide work carried out by the aforementioned students under my guidance and supervision.

Guide

Project Review Committee

HOD,CSE



DECLARATION

Date:

We hereby declare that the work presented in the field project titled “**ATTENDANCE MANAGEMENT SYSTEM**” is the result of our own efforts and investigations.

This project is being submitted under the supervision of **Mr.T.Narasimha Rao,Assistant Professor,CSE** in partial fulfillment of the requirements for the Bachelor of Technology (B.Tech.) degree in Computer Science and Engineering at the Department of Computer Science and Engineering, Vignan’s Foundation for Science, Technology and Research (Deemed to be University), Vadlamudi, Guntur, Andhra Pradesh, India.

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“Attendance Management System”

1. Introduction:

An **Attendance Management System (AMS)** is a software solution designed to efficiently track and manage attendance in various environments, such as schools, workplaces, and events. The system automates the process of recording when individuals arrive or leave, offering real-time data and eliminating the need for manual logging. AMS can integrate with other systems like payroll and student management, providing a comprehensive approach to attendance tracking. It often incorporates advanced technologies such as biometrics, RFID, or QR codes for secure and accurate tracking. This system generates automated reports, helping administrators monitor trends and make data-driven decisions. Additionally, AMS improves efficiency by saving time on attendance management, reduces human error, and ensures compliance with organizational policies. Its mobile accessibility and user-friendly interface make it easy for both administrators and users to access attendance records anytime, anywhere.

1.1. Problem Definitions:

The **Problem Definitions** in the context of an **Attendance Management System (AMS)** typically refer to the challenges or inefficiencies that such a system aims to address.

1. **Manual Attendance Tracking:** Traditionally, attendance is taken manually, which is time-consuming and prone to human error, such as incorrect marking or lost records.
2. **Time Wastage:** In schools or workplaces, manually recording attendance wastes valuable time that could be better spent on more productive activities.
3. **Lack of Real-Time Monitoring:** Traditional attendance methods do not allow real-time tracking, making it difficult for administrators or managers to get up-to-date information on attendance trends.
4. **Absence Management:** Managing absenteeism can be challenging without a proper system to track and handle leave requests, making it harder to understand the reasons behind frequent absences.
5. **Scalability Issues:** As organizations grow, managing attendance manually becomes more difficult. An AMS is essential to scale efficiently and handle large volumes of data without errors

1.2. Existing System:

1. User Authentication:

- Secure login for employees, administrators, and potentially supervisors.
- Role-based access control to manage permissions.

2. Attendance Tracking:

- Clock-in/clock-out functionality (web-based, mobile, or integrated with hardware like RFID readers).
- Time tracking with timestamps.
- Manual attendance adjustments (with audit trails).
- Support for different work schedules and shifts.

3. Leave Management:

- Leave application and approval workflows.
- Tracking of various leave types (vacation, sick leave, etc.).
- Leave balance calculations.

4. Reporting and Analytics:

- Attendance reports (daily, weekly, monthly).
- Leave reports.
- Employee attendance summaries.
- Data export capabilities (e.g., CSV, Excel).
- Graphical representation of data.

5. Integration:

- Integration with payroll systems.
- Integration with HR information systems (HRIS).
- API access for custom integrations.

6. Notifications and Alerts:

- Email or SMS notifications for attendance irregularities.
- Leave approval/rejection notifications.
- Reminders for clocking in/out.

7. User Interface:

- Responsive design to work on various devices.
- Clear and intuitive interface

1.3. Proposed System:

1. Core Features (Attendance-Focused):

- **User Management:**

- Role-based access control (Admin, Supervisor, Employee).
- Secure user authentication (username/password, potentially OAuth 2.0).
- Employee profiles (personal information, department, job title).
- **Attendance Tracking:**
 - Web-based clock-in/clock-out with timestamping.
 - Geolocation tracking (optional, with user consent) to verify location.
 - Manual attendance adjustments (with admin approval and audit logs).
 - Support for various shift schedules and flexible work hours.
 - Integration with QR code for clocking in/out.
- **Reporting and Analytics:**
 - Customizable attendance reports (daily, weekly, monthly).
 - Employee attendance summaries.
 - Graphical dashboards for visualizing attendance data.
 - Export reports in CSV, Excel, or PDF formats.
- **Notifications and Alerts:**
 - Email and/or in-app notifications for attendance irregularities.
 - Reminders for clocking in/out.
- **System Administration:**
 - System configuration.
 - User role and permissions management.
 - Audit logs for all actions within the system.

1.4. Literature Review:

- **Manual Systems:** Early attendance management systems relied heavily on manual processes, such as taking roll calls or using paper-based attendance sheets (Tiwari & Ranjan, 2015). These systems were often inaccurate, time-consuming, and prone to human error.
- **Cloud-based and Web-based AMS**
- **Web-based Systems:** Web-based AMS solutions have gained popularity due to their flexibility, cost-effectiveness, and ease of use. These systems allow users to mark attendance from any location using a browser or mobile device. According to research by Sharma and Verma (2020), web-based systems reduce the cost of hardware.

2.System Requirements:

2.1. Hardware Requirements:

Absolutely! Let's break down the provided requirements into distinct hardware and software components.

Component	Specification
Processor	Quad-core or higher (e.g., Intel Core i5/i7, AMD Ryzen 5/7)
RAM	Minimum 8 GB (Recommended: 16 GB)
Storage	SSD, Minimum 500 GB (Recommended: 1 TB or cloud storage like AWS, GCP, Azure)
Network	High-speed internet (≥ 1 Mbps), Stable LAN for local hosting
Desktop/Laptop	Intel Core i3 or higher, 4 GB RAM (8 GB recommended)

Software Requirements:

Component	Specification
Web Server Software	Apache HTTP Server or Nginx
Database Server	MySQL (relational SQL database)
Frontend Development	HTML5,CSS3,JavaScript, React.js / Angular / Vue.js
Supported Browsers	Latest version of Google Chrome, Microsoft Edge
Backend Development	Python (Django/Flask), PHP (Laravel), Ruby on Rails, Java (Spring Boot)

2.2. Software Requirements Specification (SRS):

- **Purpose:**
 - This document outlines the software requirements for the Attendance Management System (AMS), a web-based application designed to efficiently track and manage employee attendance.
- **Scope:**
 - This system will provide features for employee clock-in/clock-out, attendance reporting, and system administration. It will *not* include leave management.

Definitions, Acronyms, and Abbreviations:

- AMS: Attendance Management System
- API: Application Programming Interface

- JWT: JSON Web Token
- QR: Quick Response
- **References:**
 - (List any relevant documents, such as existing system documentation, user manuals, etc.)
- **Overview:**
 - This document is organized into sections covering functional requirements, non-functional requirements, and interface requirements.

2. Overall Description

- **Product Perspective:**
 - The AMS will be a standalone web application accessible through standard web browsers. It may integrate with optional hardware such as QR code scanners.
- **Product Functions:**
 - The system will provide:
 - Employee clock-in/clock-out functionality.
 - Attendance tracking and reporting.
 - System administration for user management and configuration.

User Classes and Characteristics:

- **Administrator:** Manages the system, users, and configurations.
- **Supervisor:** Can view reports of the employees they supervise.
- **Employee:** Clocks in/out, views personal attendance records.
- **Operating Environment:**
 - The system will operate in a web-based environment, accessible via modern web browsers (Chrome, Edge, Firefox).
 - Server side should run on a linux based operating system.

3.System Design:

3.1 Modules of System:

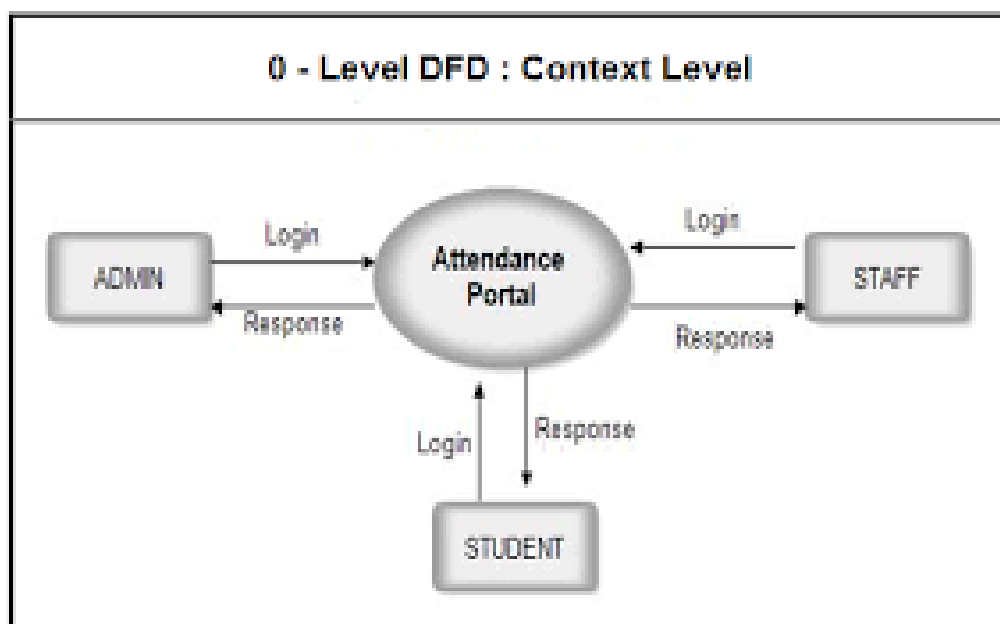
1. Architecture:

- **Multi-Tier Architecture:**
 - We'll use a three-tier architecture for clear separation of concerns:
 - **Presentation Tier (Front-End):**
 - React.js application for the user interface.
 - Handles user interactions and displays data.
 - **Application Tier (Back-End):**
 - Node.js with Express.js for the API.
 - Handles business logic, data processing, and API requests.
 - **Data Tier (Database):**
 - PostgreSQL database for storing attendance data.
- **RESTful API:**
 - The back-end will expose a RESTful API for communication between the front-end and the database.
 - This allows for flexibility and potential integration with other systems.
- **Authentication and Authorization:**
 - JWT (JSON Web Tokens) will be used for secure authentication.
 - Role-based access control will be implemented to manage user permissions.

3.2.UML Diagrams:

1.Use Case Diagram:

The **Use Case Diagram** provides a high-level overview of the system's functionalities and the interactions between different user roles and the system. For the AMS, the use case diagram will illustrate the major roles (Admin, Manager, Employee/Student) and the tasks they perform.

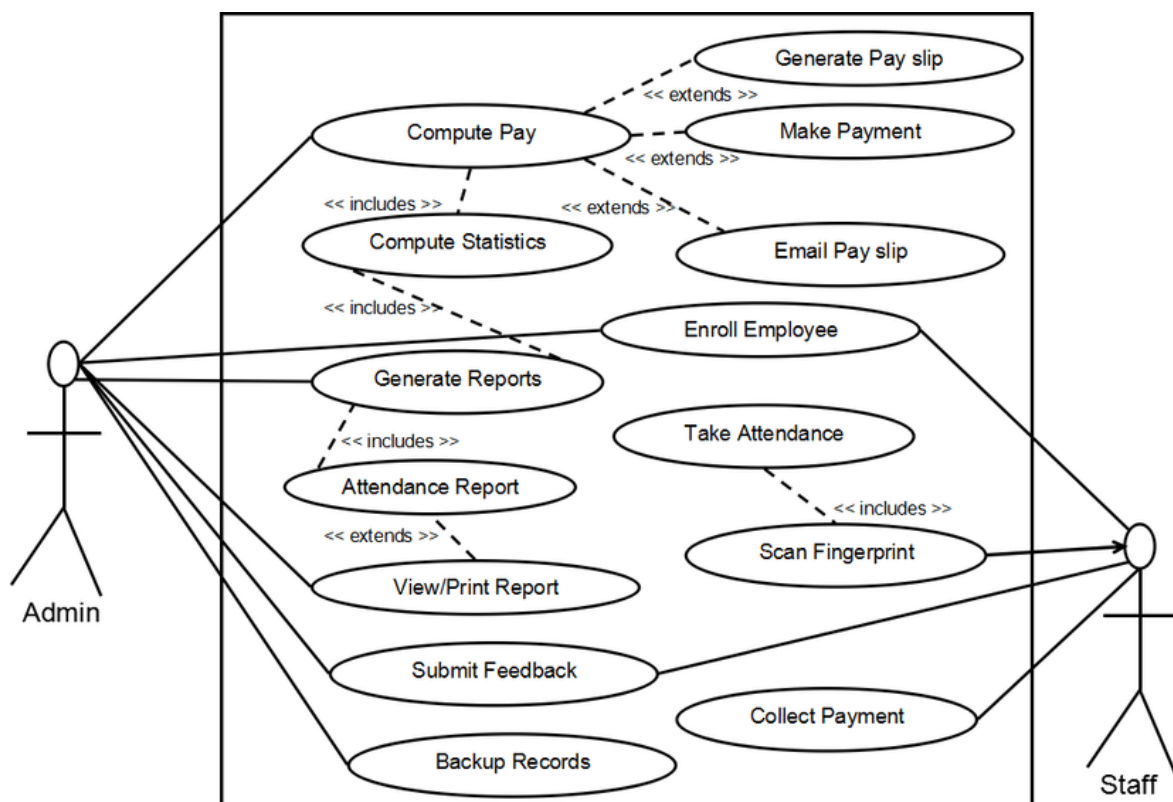


Use Case Diagram Overview

- **Actors:**
 - Admin
 - Manager
 - Employee/Student
- **Use Cases:**
 - **Login/Logout** (performed by all roles)
 - **Mark Attendance** (Student)
 - **View Reports** (Admin, Manager)
 - **Generate Reports** (Admin)
 - **Manage Users** (Admin)
 - **Configure System** (Admin)

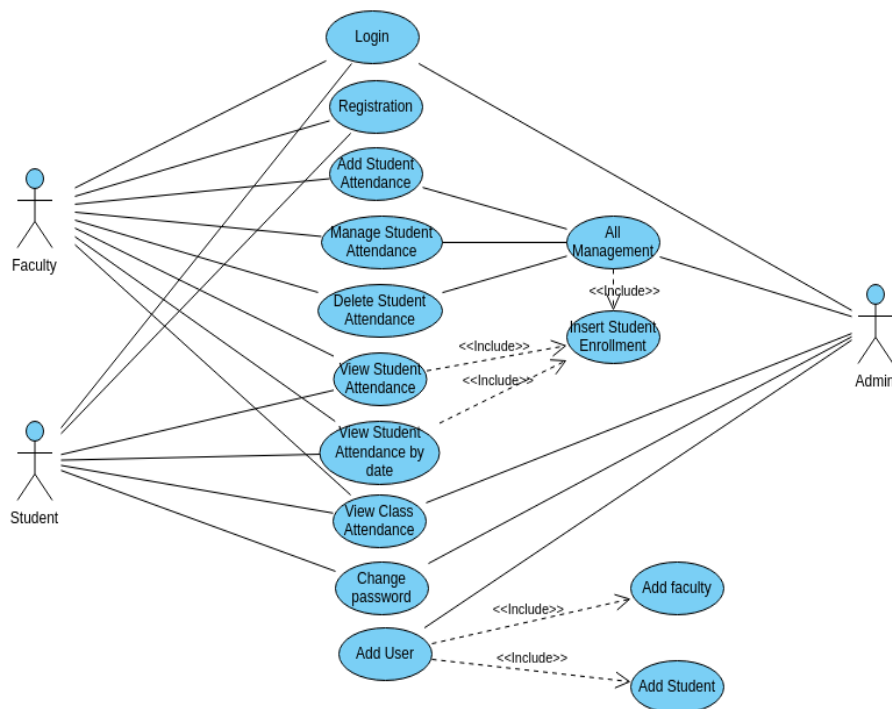
2. Class Diagram:

The **Class Diagram** shows the structure of the system by representing the classes, their attributes, methods, and relationships between them.



Class Diagram Overview

- **Classes:**
 - **User** (base class)
 - **Employee / Student** (inherits from User)
 - **Admin** (inherits from User)
 - **Attendance**
 - **Leave**



The **Activity Diagram** illustrates the flow of activities and operations in a specific process. For example, an activity diagram for **applying for leave** would show the series of steps from requesting leave to the approval or rejection.

4.Implementation:

4.1. Sample Code:

Index :

```
<!DOCTYPE html>

<html>

<head>

<title>Login - AMS</title>

<link rel="icon" href="https://img.icons8.com/stencil/32/checkmark.png" type="image/x-
icon">

<link rel="preconnect" href="https://fonts.googleapis.com">

<link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>

<link href="https://fonts.googleapis.com/css2?family=Sigmar&display=swap"
rel="stylesheet">

<link
href="https://fonts.googleapis.com/css2?family=Poppins:wght@400;600&family=Great+Vib
es&display=swap" rel="stylesheet">

<link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.0.0-
beta3/css/all.min.css">

<style>

body {

overflow: hidden;

height: 100%;

margin: 0;

padding:0;

font-family: 'Poppins', sans-serif;

background: url('https://i.ibb.co/ccrLBv38/123.png') no-repeat center center fixed;

background-size: cover;

display: flex;

justify-content: center;

align-items: center;
```

```
height: 100vh;
margin: 0;
position: relative;
flex-direction: column;
}

.header {
position: absolute;
top: 0px;
text-align: center;
width: 100%;
font-family: "Sigmar", serif;
font-size: 45px;
color: #096bf3;
display: flex;
align-items: center;
justify-content: center;
background-color: rgba(255, 255, 255, 0.8);
box-shadow: 0 4px 8px rgba(0, 0, 0, 0.2);
padding: 10px;
border-radius: 10px;
margin-right: 10px;
font-size: 45px;
color: #2f7cf8;
}

.header i {
margin-right: 10px;
font-size: 45px;

background: linear-gradient(45deg, #ff6b6b, #f06595, #cc5de8, #845ef7, #5c7cf8, #339af0,
#22b8cf, #20c997, #51cf66, #94d82d, #fcc419, #ff922b);
```

```
-webkit-background-clip: text;
background-clip: text;
-webkit-text-fill-color: transparent;
margin-right: 10px;
}
.role-selector {
position: absolute;
top: 125px;
left: 25px;
width: 70px;
height: 90px;
border-radius: 10px;
background-color: rgba(255, 255, 255, 0.8);
display: flex;
flex-direction: column;
justify-content: center;
align-items: center;
cursor: pointer;
transition: background-color 0.3s;
border: 3px solid #007bff;
}
.role-selector:hover {
background-color: rgba(255, 255, 255, 1);
}
.role-selector i {
font-size: 36px;
color: #007bff;
}
.role-selector span {
```

```
font-size: 14px;
color: #007bff;
margin-top: 5px;
}

.divider-line {
position: absolute;
top: 120px;
left: 120px;
width: 2px;
height: calc(100vh - 200px);
background-color: #007bff;
}

.role-options {
display: none;
position: absolute;
top: 230px;
left: 10px;
width: 100px;
flex-direction: column;
align-items: center;
}

.role-options button {
width: 60px;
height: 60px;
margin-bottom: 10px;
border-radius: 10px;
border: none;
background-color: rgba(255, 255, 255, 0.8);
cursor: pointer;
```



```
font-size: 16px;
display: flex;
flex-direction: column;
justify-content: center;
align-items: center;
transition: background-color 0.3s, box-shadow 0.3s;
box-shadow: 0 4px 8px rgba(0, 0, 0, 0.2);
}
.role-options button:hover {
background-color: rgba(0, 0, 0, 0.1);
box-shadow: 0 8px 16px rgba(0, 0, 0, 0.3);
}
.role-options button i {
font-size: 36px;
color: #196bc2;
}
.role-options button span {
font-size: 14px;
color: #007bff;
margin-top: 5px;
}
.login-container {
display: none;
background: rgba(255, 255, 255, 0.9);
padding: 50px;
border: 2px solid #007bff;
border-radius: 45px;
box-shadow: 0 4px 15px rgba(0, 0, 0, 0.2);
text-align: center;
```

```
}  
.login-box {  
max-width: 400px;  
margin: auto;  
}  
.login-box h1 {  
margin-bottom: 20px;  
font-weight: 600;  
color: #333;  
}  
.title {  
font-family: 'Great Vibes', cursive;  
font-size: 36px;  
color: #333;  
}  
.form-group {  
margin-bottom: 20px;  
text-align: left;  
}  
.form-group label {  
display: block;  
margin-bottom: 5px;  
font-weight: 600;  
color: #555;  
}  
.form-group input,  
.form-group select {  
width: 100%;  
padding: 12px;
```

```
border: 1px solid #ccc;
border-radius: 5px;
font-size: 16px;
}

.login-button {
width: 100%;
padding: 12px;
background-color: #007bff;
color: white;
border: none;
border-radius: 25px;
cursor: pointer;
font-weight: 600;
font-size: 16px;
display: flex;
align-items: center;
justify-content: center;
}

.login-button i {
margin-right: 8px;
}

.login-button:hover {
background-color: #0056b3;
}

.slide-show-container {
position: absolute;
top: 140px;
right: 20px;
width: 300px;
```

```
height: 200px;
border-radius: 15px;
box-shadow: 0 4px 15px rgba(224, 216, 216, 0.896);
overflow: hidden;
}

.slideshow {
position: relative;
width: 100%;
height: 100%;
animation: slide 15s infinite;
}

.slideshow img {
position: absolute;
width: 100%;
height: 100%;
border-radius: 15px;
opacity: 0;
transition: opacity 5s ease-in-out;
}

.slideshow img.active {
opacity: 1;
}

footer {
position: absolute;
bottom: 10px;
width: 100%;
text-align: center;
font-size: 14px;
color: #b9b5b5;
```

```

}

</style>

</head>

<body>

<div class="header"><i class="fas fa-calendar-check"></i>&nbsp;&nbsp;&nbsp;Attendance
Management System</div>

<div class="role-selector" onclick="showRoleOptionsPermanently()"
onclick="toggleRoleOptions()">

<i class="fas fa-user-tag"></i>

<span>Login</span>

</div>

<div class="divider-line"></div>

<div class="role-options" id="roleOptions">

<button onclick="selectRole('admin')">

<i class="fas fa-user-shield"></i>

<span>Admin</span>

</button>

<button onclick="selectRole('faculty')">

<i class="fas fa-chalkboard-teacher"></i>

<span>Faculty</span>

</button>

<button onclick="selectRole('student')">

<i class="fas fa-user-graduate"></i>

<span>Student</span>

</button>

</div>

<div class="login-container" id="loginContainer">

<div class="login-box">

<h1 class="title">Login</h1>

<form id="loginForm" onsubmit="handleLogin(event)">

```

```

<input type="hidden" id="role" name="role" required>
<div class="form-group">
<label for="username"><i class="fas fa-user"></i> Username</label>
<input type="text" id="username" name="username" readonly>
</div>
<div class="form-group">
<label for="password"><i class="fas fa-lock"></i> Password</label>
<input type="password" id="password" name="password" required>
</div>
<button type="submit" class="login-button"><i class="fas fa-sign-in-alt"></i>
Login</button>
</form>
</div>
</div>
<div class="slideshow-container">
<div class="slideshow">



</div>
</div>
<footer>
<p>&copy; 2025 ALL RIGHTS RESERVED</p>
</footer>
<script type="module">
import { initializeApp } from "https://www.gstatic.com/firebasejs/11.3.1/firebase-app.js";
import { getDatabase, ref, get, set, child } from
"https://www.gstatic.com/firebasejs/11.3.1/firebase-database.js";
const firebaseConfig = {
apiKey: "AIzaSyDHD1hv7mERPzWNkStnJ9eKI7HquS05Deo",

```

```

authDomain: "attendance-management-system-v.firebaseio.com",
projectId: "attendance-management-system-v",
storageBucket: "attendance-management-system-v.appspot.com",
messagingSenderId: "446562547736",
appId: "1:446562547736:web:85dcee5d92ea727acf6b6d",
measurementId: "G-VYFNPC5XFL"
};

const app = initializeApp(firebaseConfig);
const database = getDatabase(app);
async function getData(path) {
const dbRef = ref(database);
const snapshot = await get(child(dbRef, path));
return snapshot.exists() ? snapshot.val() : null;
}
async function setData(path, data) {
const dbRef = ref(database, path);
await set(dbRef, data);
}

let roleOptionsVisiblePermanently = false;
function toggleRoleOptions() {
const roleOptions = document.getElementById('roleOptions');
if (!roleOptionsVisiblePermanently) {
roleOptions.style.display = roleOptions.style.display === 'none' ? 'flex' : 'none';
}
}

function showRoleOptionsPermanently() {
roleOptionsVisiblePermanently = true;
document.getElementById('roleOptions').style.display = 'flex';
}

```

```

function selectRole(role) {
  document.getElementById('role').value = role;
  document.getElementById('username').readOnly = role === 'admin';
  document.getElementById('username').placeholder = role === 'admin' ? 'Admin' : (role ===
'student' ? 'Enter your roll number' : 'Enter your ID');
  document.getElementById('loginContainer').style.display = 'block';
}

async function handleLogin(event) {
  event.preventDefault();
  const role = document.getElementById('role').value;
  const username = document.getElementById('username').value;
  const password = document.getElementById('password').value;
  console.log(`Role: ${role}, Username: ${username}, Password: ${password}`);
  if (role === 'student') {
    const students = await getData('students') || [];
    const student = students.find(student => student.rollNumber === username &&
student.rollNumber === password);
    if (student) {
      alert('Student login successful!');
      sessionStorage.setItem('authenticated', 'true');
      sessionStorage.setItem('role', 'student');
      window.location.href = 'student_dashboard.html';
    } else {
      alert('Invalid roll number or password.');
```



```
alert('Faculty login successful!');
sessionStorage.setItem('authenticated', 'true');
sessionStorage.setItem('role', 'faculty');
window.location.href = 'faculty_dashboard.html';
} else {
alert('Invalid faculty ID or password.');
```

```
}
} else if (role === 'admin') {
const adminPassword = await getData('admin/password');
if (password === adminPassword) {
alert('Admin login successful!');
sessionStorage.setItem('authenticated', 'true');
sessionStorage.setItem('role', 'admin');
window.location.href = 'admin_dashboard.html';
} else {
alert('Invalid admin credentials.');
```

```
}
}
}

document.addEventListener('DOMContentLoaded', () => {
const slideshowImages = document.querySelectorAll('.slideshow img');
let currentIndex = 0;
setInterval(() => {
slideshowImages[currentIndex].classList.remove('active');
currentIndex = (currentIndex + 1) % slideshowImages.length;
slideshowImages[currentIndex].classList.add('active');
```

```
}, 5000);
});
window.toggleRoleOptions = toggleRoleOptions;
```

```
window.showRoleOptionsPermanently = showRoleOptionsPermanently;
window.selectRole = selectRole;
window.handleLogin = handleLogin;
document.addEventListener('contextmenu', function(event) {
event.preventDefault();
});
</script>
</body>
</html>
```

Admin Dashboard :

```
<!DOCTYPE html>
<html>
<head>
<title>Admin Dashboard - AMS</title>
<link rel="icon" href="https://img.icons8.com/stencil/32/checkmark.png" type="image/x-
icon">
<link href="https://fonts.googleapis.com/css2?family=Poppins:wght@400;600&display=swa
p" rel="stylesheet">
<link rel="stylesheet" href="styles.css">
<link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.0.0-
beta3/css/all.min.css">
<style>
.back-button {
position: fixed;
top: 20px;
left: 20px;
background-color: #007bff;
color: white;
border: none;
border-radius: 50%;
```

```

width: 50px;
height: 50px;
display: flex;
justify-content: center;
align-items: center;
cursor: pointer;
box-shadow: 0 4px 8px rgba(0, 0, 0, 0.2);
font-size: 24px;
}

.back-button:hover {
background-color: #0056b3;
}

</style>

</head>

<body>

<button class="back-button" onclick="goBack()"><i class="fas fa-arrow-left"></i></button>

<div class="dashboard-container">

<h1 class="dashboard-title"><i class="fas fa-tachometer-alt"></i> Admin Dashboard</h1>

<div class="cards">

<div class="card" onclick="window.location.href='manage_users.html'">

<h2><i class="fas fa-user"></i> Manage Users</h2>

<p>View and manage all users in the system.</p>

</div>

<div class="card" onclick="window.location.href='view_attendance.html'">

<h2><i class="fas fa-calendar-check"></i> Attendance Reports</h2>

<p>Generate and view attendance reports.</p>

</div>

<div class="card" onclick="window.location.href='attendance_analysis.html'">

```

```

<h2><i class="fas fa-chart-line"></i> Attendance record </h2>

<p>Analyze attendance metrics.</p>

</div>

<div class="card" onclick="window.location.href='report_card.html'">

<h2><i class="fas fa-file-alt"></i> View Report Card</h2>

<p>View and manage report cards.</p>

</div>

</div>

</div>

</div>

<script>

document.addEventListener('DOMContentLoaded', () => {

const authenticated = sessionStorage.getItem('authenticated');

const role = sessionStorage.getItem('role');

if (!authenticated || !role) {

alert('You are not authorized to view this page. Please log in.');

```

Manage Users:

```
<!DOCTYPE html>
```

```
<html>

<head>

<title>Manage Users - AMS</title>

<link rel="icon" href="https://img.icons8.com/stencil/32/checkmark.png" type="image/x-
icon">

<link
href="https://fonts.googleapis.com/css2?family=Poppins:wght@400;600&display=swap"
rel="stylesheet">

<link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.0.0-
beta3/css/all.min.css">

<style>

body {

font-family: 'Poppins', sans-serif;

background: #f0f0f0;

display: flex;

justify-content: center;

align-items: center;

height: 100vh;

margin: 0;

flex-direction: column;

}

.manage-users-container {

background: #fff;

padding: 30px;

border-radius: 15px;

box-shadow: 0 4px 15px rgba(0, 0, 0, 0.2);

width: 90%;

max-width: 1200px;

text-align: center;

display: flex;
```

```
justify-content: space-between;
gap: 20px;
box-sizing: border-box;
height: 90vh;
overflow-y: hidden;
}

.form-container {
width: 48%;
padding: 20px;
border-radius: 10px;
box-shadow: 0 4px 10px rgba(0, 0, 0, 0.1);
background: #f9f9f9;
box-sizing: border-box;
position: sticky;
top: 0;
height: 100%;
overflow-y: auto;
}

.manage-users-container h1 {
margin-bottom: 20px;
font-weight: 600;
color: #333;
}

.form-group {
margin-bottom: 20px;
text-align: left;
}

.form-group label {
display: block;
```

```
margin-bottom: 5px;
font-weight: 600;
color: #555;
}

.form-group input,
.form-group select {
width: 100%;
padding: 12px;
border: 1px solid #ccc;
border-radius: 5px;
font-size: 16px;
background: #fff;
box-shadow: inset 0 1px 3px rgba(0, 0, 0, 0.1);
box-sizing: border-box;
}

.submit-button {
width: 100%;
padding: 12px;
background-color: #007bff;
color: white;
border: none;
border-radius: 5px;
cursor: pointer;
font-weight: 600;
font-size: 16px;
display: flex;
align-items: center;
justify-content: center;
transition: background-color 0.3s ease;
```

```
}  
.submit-button i {  
margin-right: 8px;  
}  
.submit-button:hover {  
background-color: #0056b3;  
}  
.user-list {  
margin-top: 20px;  
text-align: left;  
max-height: 300px;  
overflow-y: auto;  
}  
.user-list h2 {  
font-size: 18px;  
font-weight: 600;  
color: #333;  
}  
.user-list ul {  
list-style-type: none;  
padding: 0;  
}  
.user-list li {  
margin-bottom: 10px;  
display: flex;  
justify-content: space-between;  
align-items: center;  
padding: 10px;  
background: #fff;
```



```
border-radius: 5px;
box-shadow: 0 2px 5px rgba(0, 0, 0, 0.1);
}
.user-list button {
background-color: #dc3545;
color: white;
border: none;
border-radius: 5px;
cursor: pointer;
padding: 5px 10px;
transition: background-color 0.3s ease;
}
.user-list button:hover {
background-color: #c82333;
}
.back-button {
position: fixed;
top: 20px;
left: 20px;
background-color: #007bff;
color: white;
border: none;
border-radius: 50%;
width: 50px;
height: 50px;
display: flex;
justify-content: center;
align-items: center;
cursor: pointer;
```

```

    box-shadow: 0 4px 8px rgba(0, 0, 0, 0.2);
    font-size: 24px;
    z-index: 1000;
}

.back-button:hover {
    background-color: #0056b3;
}

</style>
</head>
<body>
    <button class="back-button" onclick="goBack()"><i class="fas fa-arrow-left"></i></button>
    <div class="manage-users-container">
        <div class="form-container">
            <h1>Manage Students</h1>
            <form id="studentForm">
                <div class="form-group">
                    <label for="studentName">Student Name</label>
                    <input type="text" id="studentName" name="studentName" required>
                </div>
                <div class="form-group">
                    <label for="rollNumber">Roll Number</label>
                    <div style="display: flex; align-items: center;">
                        <span style="flex-shrink: 0;">251FA04</span>
                        <input type="text" id="rollNumber" name="rollNumber" maxlength="3" required
                        style="flex-grow: 1; margin-left: 5px;">
                    </div>
                </div>
                <div class="form-group">
                    <label for="section">Section</label>

```

```

<select id="section" name="section" required>
<option value="A">A</option>
<option value="B">B</option>
<option value="C">C</option>
<option value="D">D</option>
<option value="E">E</option>
</select>

</div>

<button type="submit" class="submit-button"><i class="fas fa-save"></i> Save
Student</button>

</form>

<div class="user-list">

<h2>Student List</h2>

<div class="form-group" style="display: flex; gap: 10px;">

<div style="flex-grow: 1;">

<label for="filterSection">Filter by Section</label>

<select id="filterSection" name="filterSection" style="width: 100%;"
onchange="filterStudents()">

<option value="">All Sections</option>

<option value="A">A</option>

<option value="B">B</option>

<option value="C">C</option>

<option value="D">D</option>

<option value="E">E</option>

</select>

</div>

<div style="flex-grow: 1;">

<label for="searchRollNumber">Search by Roll Number</label>

<input type="text" id="searchRollNumber" name="searchRollNumber" placeholder="Enter
Roll Number" oninput="filterStudents()">

```

```
</div>
</div>
<ul id="studentList"></ul>
</div>
</div>
<div class="form-container">
<h1>Manage Faculty</h1>
<form id="facultyForm">
<div class="form-group">
<label for="facultyName">Faculty Name</label>
<input type="text" id="facultyName" name="facultyName" required>
</div>
<div class="form-group">
<label for="facultyId">Faculty ID</label>
<input type="text" id="facultyId" name="facultyId" required>
</div>
<button type="submit" class="submit-button"><i class="fas fa-save"></i> Save
Faculty</button>
</form>
<div class="user-list">
<h2>Faculty List</h2>
<ul id="facultyList"></ul>
</div>
</div>
</div>
<script>
function goBack() {
window.history.back();
}
```

```

</script>

<script type="module">

import { initializeApp } from "https://www.gstatic.com/firebasejs/11.3.1/firebase-app.js";

import { getDatabase, ref, get, set, child, remove } from
"https://www.gstatic.com/firebasejs/11.3.1/firebase-database.js";

const firebaseConfig = {

apiKey: "AIzaSyDHD1hv7mERPzWNkStnJ9eKI7HquS05Deo",

authDomain: "attendance-management-system-v.firebaseio.com",

projectId: "attendance-management-system-v",

storageBucket: "attendance-management-system-v.appspot.com",

messagingSenderId: "446562547736",

appId: "1:446562547736:web:85dcee5d92ea727acf6b6d",

measurementId: "G-VYFNPC5XFL"

};

const app = initializeApp(firebaseConfig);

const database = getDatabase(app);

async function getData(path) {

const dbRef = ref(database);

const snapshot = await get(child(dbRef, path));

return snapshot.exists() ? snapshot.val() : [];

}

async function setData(path, data) {

const dbRef = ref(database, path);

await set(dbRef, data);

}

async function removeData(path) {

const dbRef = ref(database, path);

await remove(dbRef);

}

```

```

window.removeStudent = removeStudent;
window.removeFaculty = removeFaculty;
window.filterStudents = filterStudents;
document.getElementById('studentForm').addEventListener('submit', async function(event) {
event.preventDefault();
const studentName = document.getElementById('studentName').value;
const rollNumberSuffix = document.getElementById('rollNumber').value;
const rollNumber = `251FA04${rollNumberSuffix}`;
const section = document.getElementById('section').value;
const students = await getData('students');
const existingStudentIndex = students.findIndex(student => student.rollNumber ===
rollNumber);
if (existingStudentIndex !== -1) {
students[existingStudentIndex] = { studentName, rollNumber, section };
} else {
students.push({ studentName, rollNumber, section });
}
students.sort((a, b) => a.rollNumber.localeCompare(b.rollNumber));
await setData('students', students);
document.getElementById('studentForm').reset();
displayStudents();
});
document.getElementById('facultyForm').addEventListener('submit', async function(event) {
event.preventDefault();
const facultyName = document.getElementById('facultyName').value;
const facultyId = document.getElementById('facultyId').value;
const faculties = await getData('faculties');
const existingFacultyIndex = faculties.findIndex(faculty => faculty.facultyId === facultyId);
if (existingFacultyIndex !== -1) {

```

```

faculties[existingFacultyIndex] = { facultyName, facultyId };
} else {
faculties.push({ facultyName, facultyId });
}
await setData('faculties', faculties);
document.getElementById('facultyForm').reset();
displayFaculties();
});
async function displayStudents() {
const students = await getData('students');
const studentList = document.getElementById('studentList');
studentList.innerHTML = "";
students.forEach(student => {
const li = document.createElement('li');
li.innerHTML = `
${student.studentName} (${student.rollNumber}) - Section ${student.section}
<button onclick="removeStudent('${student.rollNumber}')">Remove</button>
`;
studentList.appendChild(li);
});
}
async function displayFaculties() {
const faculties = await getData('faculties');
const facultyList = document.getElementById('facultyList');
facultyList.innerHTML = "";
faculties.forEach(faculty => {
const li = document.createElement('li');
li.innerHTML = `
${faculty.facultyName} (${faculty.facultyId})

```

```

<button onclick="removeFaculty('${faculty.facultyId}')">Remove</button>
`;
facultyList.appendChild(li);
});
}

async function removeStudent(rollNumber) {
let students = await getData('students');
students = students.filter(student => student.rollNumber !== rollNumber);
await setData('students', students);
displayStudents();
}

async function removeFaculty(facultyId) {
let faculties = await getData('faculties');
faculties = faculties.filter(faculty => faculty.facultyId !== facultyId);
await setData('faculties', faculties);
displayFaculties();
}

async function filterStudents() {
const selectedSection = document.getElementById('filterSection').value;
const searchRollNumber =
document.getElementById('searchRollNumber').value.toLowerCase();
const students = await getData('students');
const studentList = document.getElementById('studentList');
studentList.innerHTML = "";
students.forEach(student => {
if ((!selectedSection || student.section === selectedSection) &&
(!searchRollNumber || student.rollNumber.toLowerCase().includes(searchRollNumber))) {
const li = document.createElement('li');
li.innerHTML = `

```



```

    ${student.studentName} (${student.rollNumber}) - Section ${student.section}
    <button onclick="removeStudent('${student.rollNumber}')">Remove</button>
    `;
    studentList.appendChild(li);
  }
});
}

document.addEventListener('DOMContentLoaded', () => {
  displayStudents();
  displayFaculties();
});

document.addEventListener('DOMContentLoaded', () => {
  const authenticated = sessionStorage.getItem('authenticated');
  const role = sessionStorage.getItem('role');
  if (!authenticated || !role) {
    alert('You are not authorized to view this page. Please log in.');
```

window.location.href = 'index.html';

```

  }
});

document.addEventListener('contextmenu', function(event) {
  event.preventDefault();
});
</script>
</body>
</html>

```

Attendance Analysis:

```

<!DOCTYPE html>
<html lang="en">
<head>

```

```
<meta charset="UTF-8">

<meta name="viewport" content="width=device-width, initial-scale=1.0">

<title>Attendance Analysis - AMS</title>

<link rel="icon" href="https://img.icons8.com/stencil/32/checkmark.png" type="image/x-
icon">

<link
href="https://fonts.googleapis.com/css2?family=Poppins:wght@400;600&display=swap"
rel="stylesheet">

<link rel="stylesheet" href="styles.css">

<link rel="stylesheet" href="https://cdn.jsdelivr.net/npm/chart.js"></script>

<style>

.back-button {
position: fixed;
top: 20px;
left: 20px;
background-color: #007bff;
color: white;
border: none;
border-radius: 50%;
width: 50px;
height: 50px;
display: flex;
justify-content: center;
align-items: center;
cursor: pointer;
box-shadow: 0 4px 8px rgba(0, 0, 0, 0.2);
font-size: 24px;
z-index: 1000;
```

```
}  
.back-button:hover {  
background-color: #0056b3;  
}  
body {  
font-family: 'Poppins', sans-serif;  
background: #f0f0f0;  
display flex;  
justify-content: center;  
align-items: center;  
height: 100vh;  
margin: 0;  
flex-direction: column;  
}  
.analysis-container {  
background: #fff;  
padding: 30px;  
border-radius: 15px;  
box-shadow: 0 4px 15px rgba(0, 0, 0, 0.2);  
width: 90%;  
max-width: 1200px;  
text-align: center;  
}  
.analysis-container h1 {  
margin-bottom: 20px;  
font-weight: 600;  
color: #333;  
}  
.charts-container {
```

```

display: flex;
justify-content: space-around;
gap: 20px;
}
.chart-container {
width: 45%;
}
</style>
</head>
<body>
<button class="back-button" onclick="goBack()"><i class="fas fa-arrow-left"></i></button>
<div class="analysis-container">
<h1><i class="fas fa-chart-pie"></i> Attendance Analysis</h1>
<div class="charts-container">
<div class="chart-container">
<canvas id="attendancePieChart"></canvas>
</div>
<div class="chart-container">
<canvas id="attendanceLineChart"></canvas>
</div>
</div>
</div>
</div>
<script>
function goBack() {
window.history.back();
}
</script>
<script type="module">
import { initializeApp } from "https://www.gstatic.com/firebasejs/11.3.1/firebase-app.js";

```

```

import { getDatabase, ref, get, child } from
"https://www.gstatic.com/firebasejs/11.3.1/firebase-database.js";

const firebaseConfig = {
  apiKey: "AIzaSyDHD1hv7mERPzWNkStnJ9eKI7HquS05Deo",
  authDomain: "attendance-management-system-v.firebaseio.com",
  projectId: "attendance-management-system-v",
  storageBucket: "attendance-management-system-v.appspot.com",
  messagingSenderId: "446562547736",
  appId: "1:446562547736:web:85dcee5d92ea727acf6b6d",
  measurementId: "G-VYFNPC5XFL"
};

const app = initializeApp(firebaseConfig);
const database = getDatabase(app);
async function getData(path) {
  const dbRef = ref(database);
  const snapshot = await get(child(dbRef, path));
  return snapshot.exists() ? snapshot.val() : [];
}

document.addEventListener('DOMContentLoaded', async (event) => {
  const attendanceData = await getData('attendanceData');
  const totalClasses = attendanceData.length;
  const attendedClasses = attendanceData.filter(record => record.status === 'Present').length;
  const absentClasses = totalClasses - attendedClasses;
  const pieCtx = document.getElementById('attendancePieChart').getContext('2d');
  const pieChart = new Chart(pieCtx, {
    type: 'pie',
    data: {
      labels: ['Present', 'Absent'],
      datasets: [{

```

```

data: [attendedClasses, absentClasses],
backgroundColor: ['#28a745', '#ff6b6b'],
}]
},
options: {
responsive: true,
plugins: {
legend: {
position: 'top',
},
title: {
display: true,
text: 'Overall Attendance'
}
}
});

const subjects = ['DAA', 'OS', 'TOC', 'COA'];

const subjectAttendance = subjects.map(subject => {

const totalSubjectClasses = attendanceData.filter(record => record.subject ===
subject).length;

const attendedSubjectClasses = attendanceData.filter(record => record.subject === subject
&& record.status === 'Present').length;

return {
subject,
total: totalSubjectClasses,
attended: attendedSubjectClasses
};

});

const lineCtx = document.getElementById('attendanceLineChart').getContext('2d');

```

```

const lineChart = new Chart(lineCtx, {
  type: 'line',
  data: {
    labels: subjects,
    datasets: [{
      label: 'Total Classes',
      data: subjectAttendance.map(data => data.total),
      borderColor: '#007bff',
      backgroundColor: '#007bff',
      fill: false,
    }, {
      label: 'Attended Classes',
      data: subjectAttendance.map(data => data.attended),
      borderColor: '#28a745',
      backgroundColor: '#28a745',
      fill: false,
    }]
  },
  options: {
    responsive: true,
    plugins: {
      legend: {
        position: 'top',
      },
      title: {
        display: true,
        text: 'Subject-wise Attendance'
      }
    }
  }
});

```

```

    }
  });
});

document.addEventListener('DOMContentLoaded', () => {
  const authenticated = sessionStorage.getItem('authenticated');
  const role = sessionStorage.getItem('role');
  if (!authenticated || !role) {
    alert('You are not authorized to view this page. Please log in.');
```

window.location.href = 'index.html';

```

  }
});

document.addEventListener('contextmenu', function(event) {
  event.preventDefault();
});
</script>
</body>
</html>

```

Report Card:

```

<!DOCTYPE html>
<html>
<head>
<title>View Report Card - AMS</title>
<link rel="icon" href="https://img.icons8.com/stencil/32/checkmark.png" type="image/x-
icon">
<link
href="https://fonts.googleapis.com/css2?family=Poppins:wght@400;600&display=swap"
rel="stylesheet">
<link rel="stylesheet" href="styles.css">
<link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.0.0-
beta3/css/all.min.css">

```



```
<style>

body {
  font-family: 'Poppins', sans-serif;
  background: #f0f0f0;
  display: flex;
  justify-content: center;
  align-items: center;
  height: 100vh;
  margin: 0;
  flex-direction: column;
}

.report-card-container {
  background: #fff;
  padding: 30px;
  border-radius: 15px;
  box-shadow: 0 4px 15px rgba(0, 0, 0, 0.2);
  width: 90%;
  max-width: 600px;
  text-align: center;
}

.report-card-container h1 {
  margin-bottom: 20px;
  font-weight: 600;
  color: #333;
}

.form-group {
  margin-bottom: 20px;
  text-align: left;
}
```

```
.form-group label {  
display: block;  
margin-bottom: 5px;  
font-weight: 600;  
color: #555;  
}  
  
.form-group input {  
width: 100%;  
padding: 12px;  
border: 1px solid #ccc;  
border-radius: 5px;  
font-size: 16px;  
}  
  
.submit-button {  
width: 100%;  
padding: 12px;  
background-color: #007bff;  
color: white;  
border: none;  
border-radius: 5px;  
cursor: pointer;  
font-weight: 600;  
font-size: 16px;  
display: flex;  
align-items: center;  
justify-content: center;  
}  
  
.submit-button i {  
margin-right: 8px;
```

```
}  
.submit-button:hover {  
background-color: #0056b3;  
}  
.attendance-result {  
margin-top: 20px;  
font-size: 18px;  
font-weight: 600;  
color: #333;  
}  
.subject-attendance {  
margin-top: 10px;  
font-size: 16px;  
color: #555;  
}  
.back-button {  
position: fixed;  
top: 20px;  
left: 20px;  
background-color: #007bff;  
color: white;  
border: none;  
border-radius: 50%;  
width: 50px;  
height: 50px;  
display: flex;  
justify-content: center;  
align-items: center;  
cursor: pointer;
```

```

    box-shadow: 0 4px 8px rgba(0, 0, 0, 0.2);
    font-size: 24px;
    z-index: 1000;
}

.back-button:hover {
    background-color: #0056b3;
}

</style>
</head>
<body>
<button class="back-button" onclick="goBack()"><i class="fas fa-arrow-left"></i></button>
<div class="report-card-container">
<h1><i class="fas fa-file-alt"></i> View Report Card</h1>
<form id="reportCardForm">
<div class="form-group">
<label for="rollNumber">Enter Roll Number</label>
<input type="text" id="rollNumber" name="rollNumber" required>
</div>
<button type="submit" class="submit-button"><i class="fas fa-search"></i> View
Attendance</button>
</form>
<div id="attendanceResult" class="attendance-result"></div>
<div id="subjectAttendance" class="subject-attendance"></div>
</div>
<script>
function goBack() {
    window.history.back();
}
</script>

```

```

<script type="module">

import { initializeApp } from "https://www.gstatic.com/firebasejs/11.3.1/firebase-app.js";

import { getDatabase, ref, get, child } from
"https://www.gstatic.com/firebasejs/11.3.1/firebase-database.js";

const firebaseConfig = {
  apiKey: "AIzaSyDHD1hv7mERPzWNkStnJ9eKI7HquS05Deo",
  authDomain: "attendance-management-system-v.firebaseio.com",
  projectId: "attendance-management-system-v",
  storageBucket: "attendance-management-system-v.appspot.com",
  messagingSenderId: "446562547736",
  appId: "1:446562547736:web:85dcee5d92ea727acf6b6d",
  measurementId: "G-VYFNPC5XFL"
};

const app = initializeApp(firebaseConfig);
const database = getDatabase(app);

async function getData(path) {
  const dbRef = ref(database);

  const snapshot = await get(child(dbRef, path));

  return snapshot.exists() ? snapshot.val() : [];
}

document.getElementById('reportCardForm').addEventListener('submit', async
function(event) {

  event.preventDefault();

  const rollNumber = document.getElementById('rollNumber').value;

  const attendanceData = await getData('attendanceData');

  const totalClassesData = await getData('totalClassesData');

  const studentAttendanceData = attendanceData.filter(record => record.rollNumber ===
rollNumber);

  if (studentAttendanceData.length === 0) {

    document.getElementById('attendanceResult').innerText = 'Student data not found.';
  }
}

```

```

document.getElementById('subjectAttendance').innerHTML = "";

return;
}

const totalClasses = studentAttendanceData.length;

const attendedClasses = studentAttendanceData.filter(record => record.status ===
'Present').length;

const attendancePercentage = totalClasses > 0 ? (attendedClasses / totalClasses) * 100 : 0;

document.getElementById('attendanceResult').innerText = `Total Attendance Percentage:
${attendancePercentage.toFixed(2)}%`;

const subjects = ['DAA', 'OS', 'TOC', 'COA'];

let subjectAttendanceHTML = "";

subjects.forEach(subject => {

const totalSubjectClasses = totalClassesData[subject] || 0;

const attendedSubjectClasses = studentAttendanceData.filter(record => record.subject ===
subject && record.status === 'Present').length;

const subjectAttendancePercentage = totalSubjectClasses > 0 ? (attendedSubjectClasses /
totalSubjectClasses) * 100 : 0;

subjectAttendanceHTML += `<p>${subject}: ${subjectAttendancePercentage.toFixed(2)}%
(Total Classes: ${totalSubjectClasses}, Attended: ${attendedSubjectClasses})</p>`;

});

document.getElementById('subjectAttendance').innerHTML = subjectAttendanceHTML;

});

document.addEventListener('DOMContentLoaded', () => {

const authenticated = sessionStorage.getItem('authenticated');

const role = sessionStorage.getItem('role');

if (!authenticated || !role) {

alert('You are not authorized to view this page. Please log in.');
```

window.location.href = 'index.html';

```

}

});

document.addEventListener('contextmenu', function(event) {

```

```
event.preventDefault();  
});  
</script>  
</body>  
</html>
```

View Attendance:

```
<!DOCTYPE html>  
<html lang="en">  
<head>  
<meta charset="UTF-8">  
<meta name="viewport" content="width=device-width, initial-scale=1.0">  
<title>View Attendance - AMS</title>  
<link rel="icon" href="https://img.icons8.com/stencil/32/checkmark.png" type="image/x-icon">  
<link  
href="https://fonts.googleapis.com/css2?family=Poppins:wght@400;600&display=swap"  
rel="stylesheet">  
<link rel="stylesheet" href="styles.css">  
<link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.0.0-beta3/css/all.min.css">  
<style>  
body {  
font-family: 'Poppins', sans-serif;  
background: #f0f0f0;  
display: flex;  
justify-content: center;  
align-items: center;  
height: 100vh;  
margin: 0;  
flex-direction: column;
```

```
}  
  
.attendance-container {  
background: #fff;  
padding: 30px;  
border-radius: 15px;  
box-shadow: 0 4px 15px rgba(0, 0, 0, 0.2);  
width: 90%;  
max-width: 1200px;  
margin: 50px auto;  
text-align: center;  
height: auto;  
display: flex;  
flex-direction: column;  
}  
  
.attendance-container h1 {  
margin-bottom: 20px;  
font-weight: 600;  
color: #333;  
}  
  
.clear-button {  
margin-left: 20px;  
padding: 7px 7px;  
background-color: #dc3545;  
color: white;  
border: none;  
border-radius: 45px;  
cursor: pointer;  
font-weight: 600;  
font-size: 16px;
```



```
width: 220px;
}

.clear-button:hover {
background-color: #c82333;
}

.section-select-container {
margin-top: 20px;
text-align: left;
}

.table-container {
flex-grow: 1;
overflow-y: auto;
max-height: 60vh;
}

.attendance-table {
width: 100%;
border-collapse: collapse;
margin-top: 20px;
}

.attendance-table th, .attendance-table td {
border: 1px solid #ddd;
padding: 8px;
text-align: left;
}

.attendance-table th {
background-color: #007bff;
color: white;
font-weight: 600;
position: sticky;
```

```
top: 0;
z-index: 1;
}

.attendance-table tr:nth-child(even) {
background-color: #f2f2f2;
}

.attendance-table tr:hover {
background-color: #ddd;
}

.status-present {
background-color: #d4edda;
}

.status-absent {
background-color: #f8d7da;
}

.back-button {
position: fixed;
top: 20px;
left: 20px;
background-color: #007bff;
color: white;
border: none;
border-radius: 50%;
width: 50px;
height: 50px;
display: flex;
justify-content: center;
align-items: center;
cursor: pointer;
```

```

    box-shadow: 0 4px 8px rgba(0, 0, 0, 0.2);
    font-size: 24px;
    z-index: 1000;
}

.back-button:hover {
    background-color: #0056b3;
}

</style>

</head>

<body>

<button class="back-button" onclick="goBack()"><i class="fas fa-arrow-left"></i></button>

<div class="attendance-container">

<h1><i class="fas fa-calendar-check"></i> Attendance Details</h1>

<button class="clear-button" onclick="clearAttendanceData()">Clear Attendance
Data</button>

<div class="section-select-container">

<label for="sectionSelect">Select Section:</label>

<select id="sectionSelect" onchange="fetchSectionAttendance()">

<option value="">-- Select Section --</option>

<option value="A">A</option>

<option value="B">B</option>

<option value="C">C</option>

<option value="D">D</option>

<option value="E">E</option>

</select>

</div>

<div class="table-container">

<table class="attendance-table">

<thead>

```

```

<tr>
<th>Roll Number</th>
<th>Period</th>
<th>Subject</th>
<th>Date</th>
<th>Status</th>
</tr>
</thead>
<tbody id="attendanceTableBody">
</tbody>
</table>
</div>
</div>
<script>
function goBack() {
window.history.back();
}
</script>
<script type="module">
import { initializeApp } from "https://www.gstatic.com/firebasejs/11.3.1/firebase-app.js";
import { getDatabase, ref, get, query, orderByChild, equalTo, set } from
"https://www.gstatic.com/firebasejs/11.3.1/firebase-database.js";
const firebaseConfig = {
apiKey: "AIzaSyDHD1hv7mERPzWNkStnJ9eKI7HquS05Deo",
authDomain: "attendance-management-system-v.firebaseio.com",
projectId: "attendance-management-system-v",
storageBucket: "attendance-management-system-v.appspot.com",
messagingSenderId: "446562547736",
appId: "1:446562547736:web:85dcee5d92ea727acf6b6d",

```

```

measurementId: "G-VYFNPC5XFL",

databaseURL: "https://attendance-management-system-v-default-rtdb.firebaseio.com/" //
Ensure this is correct

};

const app = initializeApp(firebaseConfig);
const database = getDatabase(app);
async function fetchSectionAttendance() {
const selectedSection = document.getElementById('sectionSelect').value;
const attendanceTableBody = document.getElementById('attendanceTableBody');
attendanceTableBody.innerHTML = "";
if (selectedSection) {
try {
const attendanceRef = ref(database, 'attendanceData');
const studentsRef = ref(database, 'students');

const [attendanceSnapshot, studentsSnapshot] = await Promise.all([get(attendanceRef),
get(studentsRef)]);

if (attendanceSnapshot.exists() && studentsSnapshot.exists()) {
const attendanceData = attendanceSnapshot.val();
const studentsData = studentsSnapshot.val();

const filteredData = Object.values(attendanceData).filter(record => {

const student = Object.values(studentsData).find(student => student.rollNumber ===
record.rollNumber);

return student && student.section === selectedSection;

});

if (filteredData.length > 0) {
filteredData.forEach(record => {

const row = document.createElement('tr');

const statusClass = record.status === 'Present' ? 'status-present' : 'status-absent';

row.innerHTML = `

<td>${record.rollNumber}</td>

```

```

<td>${record.period}</td>
<td>${record.subject}</td>
<td>${record.date}</td>
<td class="${statusClass}">${record.status}</td>
attendanceTableBody.appendChild(row);
});
} else {
console.log('No attendance data found for this section.');
```

```

document.addEventListener('DOMContentLoaded', () => {
});

document.addEventListener('DOMContentLoaded', () => {
const authenticated = sessionStorage.getItem('authenticated');
const role = sessionStorage.getItem('role');
if (!authenticated || !role) {
alert('You are not authorized to view this page. Please log in.');
```

```

window.location.href = 'index.html';
}
});

document.addEventListener('contextmenu', function(event) {
event.preventDefault();
});
</script>
</body>
</html>

```

Faculty Dashboard:

```

<!DOCTYPE html>
<html>
<head>
<title>Faculty Dashboard - AMS</title>
<link rel="icon" href="https://img.icons8.com/stencil/32/checkmark.png" type="image/x-
icon">
<link rel="stylesheet" href="styles.css">
<link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.0.0-
beta3/css/all.min.css">
<style>
.back-button {
position: fixed;
top: 20px;

```

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```

left: 20px;
background-color: #007bff;
color: white;
border: none;
border-radius: 50%;
width: 50px;
height: 50px;
display: flex;
justify-content: center;
align-items: center;
cursor: pointer;
box-shadow: 0 4px 8px rgba(0, 0, 0, 0.2);
font-size: 24px;
z-index: 1000;
}

.back-button:hover {
background-color: #0056b3;
}

</style>

</head>

<body>

<button class="back-button" onclick="goBack()"><i class="fas fa-arrow-left"></i></button>

<div class="dashboard-container">

<h1 class="dashboard-title"><i class="fas fa-chalkboard-teacher"></i> Faculty
Dashboard</h1>

<div class="cards">

<div class="card" onclick="window.location.href='mark_attendance.html'">

<h2>Mark Attendance</h2>

<p>Mark attendance for students.</p>

```



```

</div>

<div class="card" onclick="window.location.href='view_attendance.html'">

<h2>View Attendance</h2>

<p>View attendance records.</p>

</div>

</div>

</div>

<script>

function goBack() {
window.history.back();
}

document.addEventListener('DOMContentLoaded', () => {
const authenticated = sessionStorage.getItem('authenticated');
const role = sessionStorage.getItem('role');
if (!authenticated || !role) {
alert('You are not authorized to view this page. Please log in.');
window.location.href = 'index.html';
}
});

document.addEventListener('contextmenu', function(event) {
event.preventDefault();
});

</script>

</body>

</html>

```

Mark Attendance:

```

<!DOCTYPE html>

<html>

<head>

```

```
<title>Mark Attendance</title>

<link rel="icon" href="https://img.icons8.com/stencil/32/checkmark.png" type="image/x-
icon">

<link
href="https://fonts.googleapis.com/css2?family=Poppins:wght@400;600&display=swap"
rel="stylesheet">

<link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.0.0-
beta3/css/all.min.css">

<style>

body {

font-family: 'Poppins', sans-serif;

background: #f0f0f0;

display: flex;

justify-content: center;

align-items: center;

height: 100vh;

margin: 0;

flex-direction: column;

}

.attendance-container {

background: #fff;

padding: 30px;

border-radius: 15px;

box-shadow: 0 4px 15px rgba(0, 0, 0, 0.2);

width: 90%;

max-width: 1200px;

text-align: center;

height: 90vh;

overflow-y: auto;

}
```

```
.attendance-container h1 {  
margin-bottom: 20px;  
font-weight: 600;  
color: #333;  
}  
  
.form-group {  
margin-bottom: 10px;  
text-align: left;  
display: inline-block;  
width: 22%;  
margin-right: 2%;  
}  
  
.form-group label {  
display: block;  
margin-bottom: 5px;  
font-weight: 600;  
color: #555;  
}  
  
.form-group select,  
.form-group input {  
width: 100%;  
padding: 8px;  
border: 1px solid #ccc;  
border-radius: 5px;  
font-size: 14px;  
}  
  
.form-group input[type="date"] {  
font-family: 'Poppins', sans-serif;  
font-size: 14px;
```

```
color: #555;
}

.attendance-button {
width: 30%;
padding: 12px;
background-color: #007bff;
color: white;
border: none;
border-radius: 5px;
cursor: pointer;
font-weight: 600;
font-size: 16px;
margin-top: 20px;
}

.attendance-button:hover {
background-color: #0056b3;
}

.roll-numbers-container {
display: none;
width: 100%;
height: 100vh;
position: absolute;
top: 0;
left: 0;
background: #fff;
padding: 30px;
box-sizing: border-box;
}

.roll-numbers {
```

```
display: grid;
grid-template-columns: repeat(8, 1fr);
gap: 10px;
height: calc(100vh - 150px);
overflow-y: auto;
}

.roll-number {
display: flex;
align-items: center;
}

.roll-number input[type="checkbox"] {
margin-right: 10px;
width: 20px;
height: 20px;
accent-color: #007bff;
}

.submit-button {
width: 30%;
padding: 12px;
background-color: #28a745;
color: white;
border: none;
border-radius: 5px;
cursor: pointer;
font-weight: 600;
font-size: 16px;
margin-top: 20px;
}

.submit-button:hover {
```

```

background-color: #218838;
}

.back-button {
position: fixed;
top: 6px;
left: 6px;
background-color: #007bff;
color: white;
border: none;
border-radius: 50%;
width: 50px;
height: 50px;
display: flex;
justify-content: center;
align-items: center;
cursor: pointer;
box-shadow: 0 4px 8px rgba(0, 0, 0, 0.2);
font-size: 24px;
z-index: 1000;
}

.back-button:hover {
background-color: #0056b3;
}
</style>
</head>
<body>
<button class="back-button" onclick="goBack()"><i class="fas fa-arrow-left"></i></button>
<div class="attendance-container" id="attendanceContainer">
<h1>Mark Attendance</h1>

```

```
<div class="form-group">
<label for="subject">Subject</label>
<select id="subject" name="subject" required>
<option value="DAA">DAA</option>
<option value="OS">OS</option>
<option value="TOC">TOC</option>
<option value="COA">COA</option>
</select>
</div>
```

```
<div class="form-group">
<label for="period">Period</label>
<select id="period" name="period" required>
<option value="1">1</option>
<option value="2">2</option>
<option value="3">3</option>
<option value="4">4</option>
<option value="5">5</option>
<option value="6">6</option>
<option value="7">7</option>
</select>
</div>
```

```
<div class="form-group">
<label for="date">Date</label>
<input type="date" id="date" name="date" required>
</div>
```

```
<div class="form-group">
<label for="section">Section</label>
<select id="section" name="section" required>
<option value="A">A</option>
```

```

<option value="B">B</option>
<option value="C">C</option>
<option value="D">D</option>
<option value="E">E</option>
</select>
</div>

<button class="attendance-button" onclick="showRollNumbers()">Enter</button>
</div>

<div id="rollNumbersContainer" class="roll-numbers-container">
<div class="roll-numbers" id="rollNumbers"></div>

<button id="submitButton" class="submit-button"
onclick="submitAttendance()">Submit</button>
</div>

<script>
function goBack() {
window.history.back();
}
</script>

<script type="module">
import { initializeApp } from "https://www.gstatic.com/firebasejs/11.3.1/firebase-app.js";
import { getDatabase, ref, get, set, child } from
"https://www.gstatic.com/firebasejs/11.3.1/firebase-database.js";
const firebaseConfig = {
apiKey: "AIzaSyDHD1hv7mERPzWNkStnJ9eKI7HquS05Deo",
authDomain: "attendance-management-system-v.firebaseio.com",
projectId: "attendance-management-system-v",
storageBucket: "attendance-management-system-v.appspot.com",
messagingSenderId: "446562547736",
appId: "1:446562547736:web:85dcee5d92ea727acf6b6d",
measurementId: "G-VYFNPC5XFL"

```



```

};

const app = initializeApp(firebaseConfig);
const database = getDatabase(app);
async function getData(path) {
  try {
    const dbRef = ref(database);
    const snapshot = await get(child(dbRef, path));
    if (snapshot.exists()) {
      console.log(`Data fetched from ${path}:`, snapshot.val());
      return snapshot.val();
    } else {
      console.log(`No data available at ${path}`);
      return [];
    }
  } catch (error) {
    console.error(`Error fetching data from ${path}:`, error);
    return [];
  }
}

async function setData(path, data) {
  try {
    const dbRef = ref(database, path);
    await set(dbRef, data);
    console.log(`Data set to ${path}:`, data);
  } catch (error) {
    console.error(`Error setting data to ${path}:`, error);
  }
}

document.addEventListener('DOMContentLoaded', (event) => {

```

```

const today = new Date().toISOString().split('T')[0];
document.getElementById('date').value = today;
});

async function showRollNumbers() {
  console.log('showRollNumbers called');
  const section = document.getElementById('section').value;
  const students = await getData('students');
  console.log('students:', students);
  const rollNumbersContainer = document.getElementById('rollNumbers');
  rollNumbersContainer.innerHTML = "";
  const sectionStudents = students.filter(student => student.section === section);
  console.log('sectionStudents:', sectionStudents);
  sectionStudents.sort((a, b) => a.rollNumber.localeCompare(b.rollNumber));
  sectionStudents.forEach(student => {
    const rollNumberDiv = document.createElement('div');
    rollNumberDiv.className = 'roll-number';
    rollNumberDiv.innerHTML = `<input type="checkbox" id="roll${student.rollNumber}"
    name="roll${student.rollNumber}" checked><label
    for="roll${student.rollNumber}">${student.rollNumber}</label>`;
    rollNumbersContainer.appendChild(rollNumberDiv);
  });

  document.getElementById('attendanceContainer').style.display = 'none';
  document.getElementById('rollNumbersContainer').style.display = 'block';
}

async function submitAttendance() {
  const subject = document.getElementById('subject').value;
  const period = document.getElementById('period').value;
  const date = document.getElementById('date').value;
  const rollNumbersContainer = document.getElementById('rollNumbers');

```

```

const checkboxes = rollNumbersContainer.querySelectorAll('input[type="checkbox"]');
const attendanceData = await getData('attendanceData');
const totalClassesData = await getData('totalClassesData');
if (!totalClassesData[subject]) {
totalClassesData[subject] = 0;
}
totalClassesData[subject] += 1;
checkboxes.forEach(checkbox => {
const rollNumber = checkbox.name.replace('roll', "");
const status = checkbox.checked ? 'Present' : 'Absent';
attendanceData.push({ rollNumber, subject, period, date, status });
});
await setData('attendanceData', attendanceData);
await setData('totalClassesData', totalClassesData);
alert('Attendance marked successfully!');
location.reload();
}
window.showRollNumbers = showRollNumbers;
window.submitAttendance = submitAttendance;
document.addEventListener('DOMContentLoaded', () => {
const authenticated = sessionStorage.getItem('authenticated');
const role = sessionStorage.getItem('role');
if (!authenticated || !role) {
alert('You are not authorized to view this page. Please log in.');
```

```

window.location.href = 'index.html';
}
});
document.addEventListener('contextmenu', function(event) {
event.preventDefault();

```

```
});
```

```
</script>
```

```
</body>
```

```
</html>
```

Student Dashboard:

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
<meta charset="UTF-8">
```

```
<meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
<title>student dashboard - AMS</title>
```

```
<link rel="icon" href="https://img.icons8.com/stencil/32/checkmark.png" type="image/x-icon">
```

```
<link rel="stylesheet" href="styles.css">
```

```
<link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.0.0-beta3/css/all.min.css">
```

```
<style>
```

```
body {
```

```
font-family: 'Poppins', sans-serif;
```

```
background: #f0f0f0;
```

```
display: flex;
```

```
justify-content: center;
```

```
align-items: center;
```

```
height: 100vh;
```

```
margin: 0;
```

```
flex-direction: column;
```

```
}
```

```
.input-container {
```

```
display: flex;
```

```
justify-content: center;
```

```
margin-bottom: 20px;
}

.dashboard-container {
background: #fff;
padding: 30px;
border-radius: 15px;
box-shadow: 0 4px 15px rgba(0, 0, 0, 0.2);
width: 90%;
max-width: 1200px;
margin: 50px auto;
text-align: center;
height: auto;
display: flex;
flex-direction: column;
}

.dashboard-title {
margin-bottom: 20px;
font-weight: 600;
color: #333;
}

.form-group {
margin-bottom: 20px;
text-align: left;
display: flex;
}

.form-group label {
display: block;
margin-bottom: 5px;
font-weight: 600;
```

```
color: #555;
}

.form-group input {
width: 75%;
padding: 12px;
height: 20px;
border: 2px solid #ccc;
border-radius: 10px;
font-size: 16px;
transition: border-color 0.3s;
}

.form-group input:focus {
border-color: #007bff;
outline: none;
}

.submit-button {
width: 15%;
padding: 5px;
height: 43px;
background-color: #007bff;
color: white;
border: none;
border-radius: 25px;
cursor: pointer;
font-weight: 600;
font-size: 16px;
margin-left: 10px;
align-items: center;
justify-content: center;
```

```
transition: background-color 0.3s;
}

.submit-button i {
margin-right: 8px;
}

.submit-button:hover {
background-color: #0056b3;
}

.student-details, .attendance-summary {
display: none;
margin-top: 20px;
background: #f9f9f9;
padding: 20px;
border-radius: 10px;
box-shadow: 0 2px 10px rgba(0, 0, 0, 0.1);
text-align: left;
}

.student-details .student-icon {
font-size: 100px;
margin-right: 20px;
color: #007bff;
}

.attendance-summary h2 {
margin-bottom: 10px;
font-weight: 600;
color: #333;
}

.attendance-summary table {
width: 100%;
```

```
border-collapse: collapse;
margin-top: 20px;
}

.attendance-summary th, .attendance-summary td {
border: 1px solid #ddd;
padding: 8px;
text-align: left;
}

.attendance-summary th {
background-color: #007bff;
color: white;
font-weight: 600;
position: sticky;
top: 0;
z-index: 1;
}

.attendance-summary tr:nth-child(even) {
background-color: #f2f2f2;
}

.attendance-summary tr:hover {
background-color: #ddd;
}

.back-button {
position: fixed;
top: 20px;
left: 20px;
background-color: #007bff;
color: white;
border: none;
```



```

border-radius: 50%;
width: 50px;
height: 50px;
display: flex;
justify-content: center;
align-items: center;
cursor: pointer;
box-shadow: 0 4px 8px rgba(0, 0, 0, 0.2);
font-size: 24px;
z-index: 1000;
}

.back-button:hover {
background-color: #0056b3;
}

</style>
</head>
<body>
<button class="back-button" onclick="goBack()"><i class="fas fa-arrow-left"></i></button>
<div class="dashboard-container">
<h1 class="dashboard-title"><i class="fas fa-user-graduate"></i> Student Dashboard</h1>
<div>
<div class="input-container">
<div class="form-group">
<label for="rollNumberInput">Enter Roll Number:</label>
<input type="text" id="rollNumberInput" placeholder="Enter Roll Number">
</div>
<button class="submit-button" onclick="fetchStudentData()">View Attendance</button>
</div>
</div>

```

```
<div class="student-details">
  <i class="fas fa-user-graduate student-icon"></i>
  <div>
    <p><strong>Name:</strong> <span id="studentName"></span></p>
    <p><strong>Roll Number:</strong> <span id="studentRollNumber"></span></p>
    <p><strong>Section:</strong> <span id="studentSection"></span></p>
  </div>
</div>

<div class="attendance-summary">
  <h2>Attendance Summary</h2>
  <table>
    <thead>
      <tr>
        <th>Subject</th>
        <th>Attendance Percentage</th>
      </tr>
    </thead>
    <tbody id="attendanceTableBody">
    </tbody>
  </table>

  <p><strong>Total Attendance Percentage:</strong> <span
id="totalAttendancePercentage"></span>%</p>
</div>
</div>
<script>
function goBack() {
  window.history.back();
}
</script>
```

```

<script type="module">

import { initializeApp } from "https://www.gstatic.com/firebasejs/11.3.1/firebase-app.js";

import { getDatabase, ref, get, query, orderByChild, equalTo
from "https://www.gstatic.com/firebasejs/11.3.1/firebase-database.js";

const firebaseConfig = {

apiKey: "AIzaSyDHD1hv7mERPzWNkStnJ9eKI7HquS05Deo",
authDomain: "attendance-management-system-v.firebaseio.com",
projectId: "attendance-management-system-v",
storageBucket: "attendance-management-system-v.appspot.com",
messagingSenderId: "446562547736",
appId: "1:446562547736:web:85dcee5d92ea727acf6b6d",
measurementId: "G-VYFNPC5XFL",

databaseURL: "https://attendance-management-system-v-default-rtdb.firebaseio.com/"
Ensure this is correct

};

const app = initializeApp(firebaseConfig);
const database = getDatabase(app);

window.fetchStudentData = async function fetchStudentData() {
const rollNumber = document.getElementById('rollNumberInput').value;

if (rollNumber) {

const studentRef = ref(database, 'students');
const studentQuery = query(studentRef);

try {

const snapshot = await get(studentQuery);

let studentFound = false;

if (snapshot.exists()) {

const studentsData = snapshot.val();

for (const key in studentsData) {

if (studentsData[key].rollNumber === rollNumber) {

const studentData = studentsData[key];

```

```

document.getElementById('studentName').innerText = studentData.studentName;
document.getElementById('studentRollNumber').innerText = rollNumber;
document.getElementById('studentSection').innerText = studentData.section;
document.querySelector('.student-details').style.display = 'block';
document.querySelector('.attendance-summary').style.display = 'block';
fetchAttendanceData(rollNumber);
studentFound = true;
break;
}
}
}
if (!studentFound) {
alert('No student found with this roll number. Please check and try again.');
```

```

console.error('No student found with this roll number.');
```

```

}
} catch (error) {
console.error('Error fetching student data:', error);
}
} else {
alert('Please enter a roll number.');
```

```

console.error('Please enter a roll number.');
```

```

}
}

async function fetchAttendanceData(rollNumber) {
const attendanceRef = ref(database, 'attendanceData');

const attendanceQuery = query(attendanceRef, orderByChild('rollNumber'),
equalTo(rollNumber));

try {
const snapshot = await get(attendanceQuery);
```

```

if (snapshot.exists()) {
  const attendanceData = snapshot.val();
  const subjectAttendance = {};
  let totalClasses = 0;
  let totalPresent = 0;
  for (const key in attendanceData) {
    const record = attendanceData[key];
    const subject = record.subject;
    if (!subjectAttendance[subject]) {
      subjectAttendance[subject] = { present: 0, total: 0 };
    }
    subjectAttendance[subject].total++;
    if (record.status === 'Present') {
      subjectAttendance[subject].present++;
    }
  }

  const tableBody = document.getElementById('attendanceTableBody');
  tableBody.innerHTML = "";
  for (const subject in subjectAttendance) {
    const presentCount = subjectAttendance[subject].present;
    const totalCount = subjectAttendance[subject].total;
    const percentage = ((presentCount / totalCount) * 100).toFixed(2);
    const row = document.createElement('tr');
    row.innerHTML = `<td>${subject}</td><td>${percentage}%</td>`;
    tableBody.appendChild(row);
  }
  for (const subject in subjectAttendance) {
    totalClasses += subjectAttendance[subject].total;
    totalPresent += subjectAttendance[subject].present;
  }
}

```

```

    }

    const totalPercentage = ((totalPresent / totalClasses) * 100).toFixed(2);

    document.getElementById('totalAttendancePercentage').innerText = totalPercentage + '%';

    } else {

    console.error('No attendance data found for this roll number.');
```

```

    }

    } catch (error) {

    console.error('Error fetching attendance data:', error);

    }

    }

    document.addEventListener('DOMContentLoaded', () => {

    const authenticated = sessionStorage.getItem('authenticated');

    const role = sessionStorage.getItem('role');

    if (!authenticated || !role) {

    alert('You are not authorized to view this page. Please log in.');
```

```

    window.location.href = 'index.html';

    }

    });

    document.addEventListener('contextmenu', function(event)

    event.preventDefault();

    });

</script>
</body>
</html>

```

Styles using CSS:

```

.dashboard-container {

padding: 20px;

max-width: 1200px;

margin: 0 auto;

```

```
}  
.dashboard-title {  
text-align: center;  
margin-bottom: 30px;  
color: #333;  
font-size: 36px;  
font-weight: 600;  
display: flex;  
align-items: center;  
justify-content: center;  
}  
.dashboard-title i {  
margin-right: 10px;  
}  
.cards {  
display: flex;  
justify-content: space-around;  
flex-wrap: wrap;  
}  
.card {  
background: #fff;  
padding: 20px;  
border: 2px solid #007bff;  
border-radius: 10px;  
box-shadow: 0 4px 15px rgba(0, 0, 0, 0.1);  
width: 40%;  
text-align: center;  
cursor: pointer;  
transition: transform 0.3s ease;
```

```
margin-bottom: 20px;
}
.card:hover {
transform: translateY(-10px);
}
.card h2 {
color: #6a11cb;
margin-bottom: 10px;
}
.card p {
color: #555;
}
```

4.2. Test Cases:

1. Test Case: User Registration

Test Case 1.1: Successful Registration

- Test Description: Verify that a new user can successfully register with valid details.
- Test Steps:
 1. Open the registration page.
 2. Enter valid details: first name, last name, email, password, and role.
 3. Click the "Register" button.
- Expected Result: The system should display a success message, and the user should be saved in the database.

Test Case 1.2: Registration with Missing Fields

- Test Description: Verify that the system prevents registration if required fields are missing.
- Test Steps:
 1. Open the registration page.
 2. Enter only partial details (e.g., missing email or password).
 3. Click the "Register" button.
- Expected Result: The system should display an error message indicating which fields are missing or invalid.

5.Results:

1.User Registration Results:

Test Case 1.1: Successful Registration

- **Expected Result:** The user is successfully registered in the database, and the system displays a message: "User registered successfully."
- **Actual Result:** The system successfully saves the user in the database, and a success message is displayed.

Test Case 1.2: Registration with Missing Fields

- **Expected Result:** The system prompts an error message indicating which fields are missing, such as: "Email is required," "Password is required."
- **Actual Result:** The system prevents submission and displays an appropriate validation error message.

2.User Login Results:

Test Case 2.1: Successful Login

- **Expected Result:** After entering valid credentials, the user is logged in and redirected to the dashboard or home page.
- **Actual Result:** The user is logged in successfully, and a JSON Web Token (JWT) is generated, allowing access to protected routes.

Test Case 2.2: Login with Empty Credentials

- **Expected Result:** If the email or password is empty, the system displays an error message: "Email and password are required."
- **Actual Result:** The system prevents login and shows the appropriate validation message.

3.Attendance Marking Results:

Test Case 3.1: Successful Attendance Marking

- **Expected Result:** The system successfully records the attendance (either "Present," "Absent," or "Late") and saves it in the database with the correct status.
- **Actual Result:** Attendance is successfully marked and saved, with a confirmation message: "Attendance marked successfully."

Test Case 3.2: Attendance Marking with Invalid User

- **Expected Result:** The system should return an error like: "User is not authenticated. Please log in."
- **Actual Result:** The system prevents the user from marking attendance without proper authentication and displays the appropriate error message.

Test Case 3.3: Multiple Attendance Marking on the Same Day

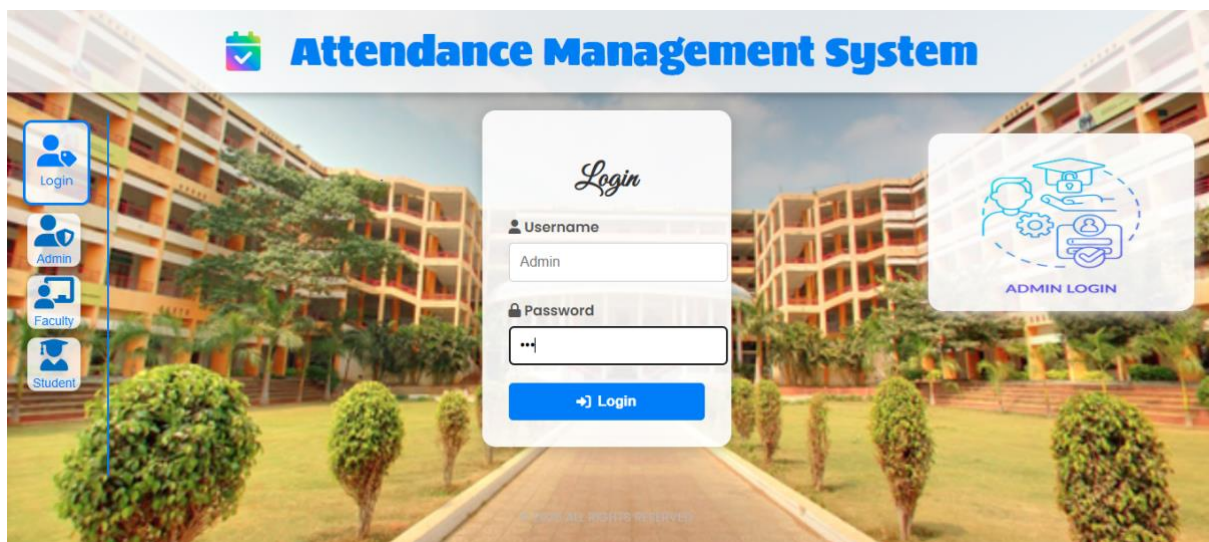
- **Expected Result:** The system should prevent the user from marking attendance more than once in a single day, displaying a message like: "Attendance has already been marked for today."
- **Actual Result:** The system prevents duplicate attendance marking and returns the expected error message.

5.1 Output Screens:

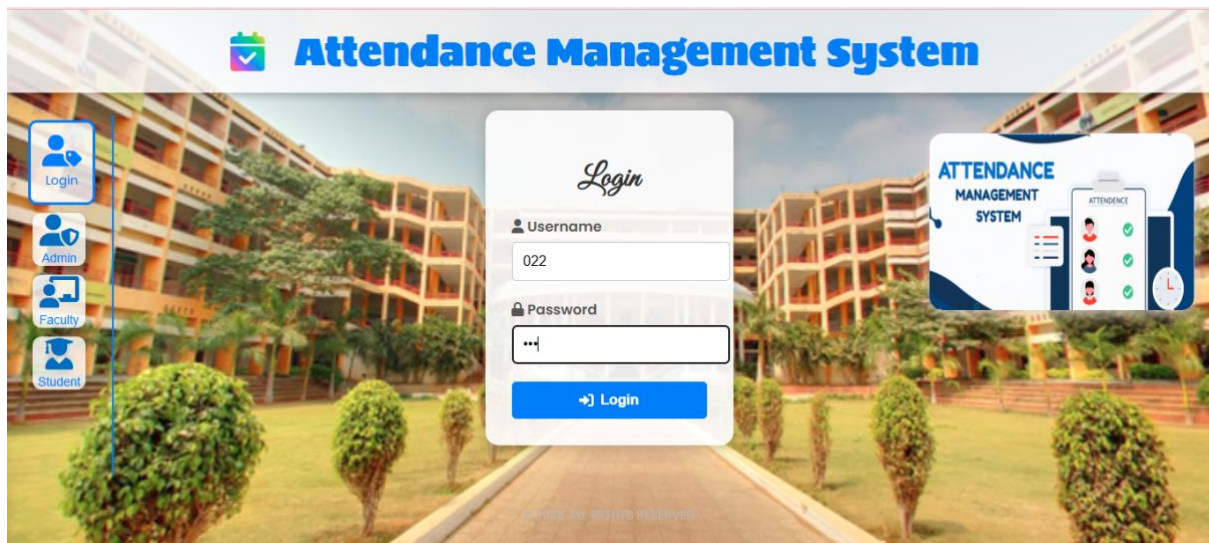
Login page:



Admin page:

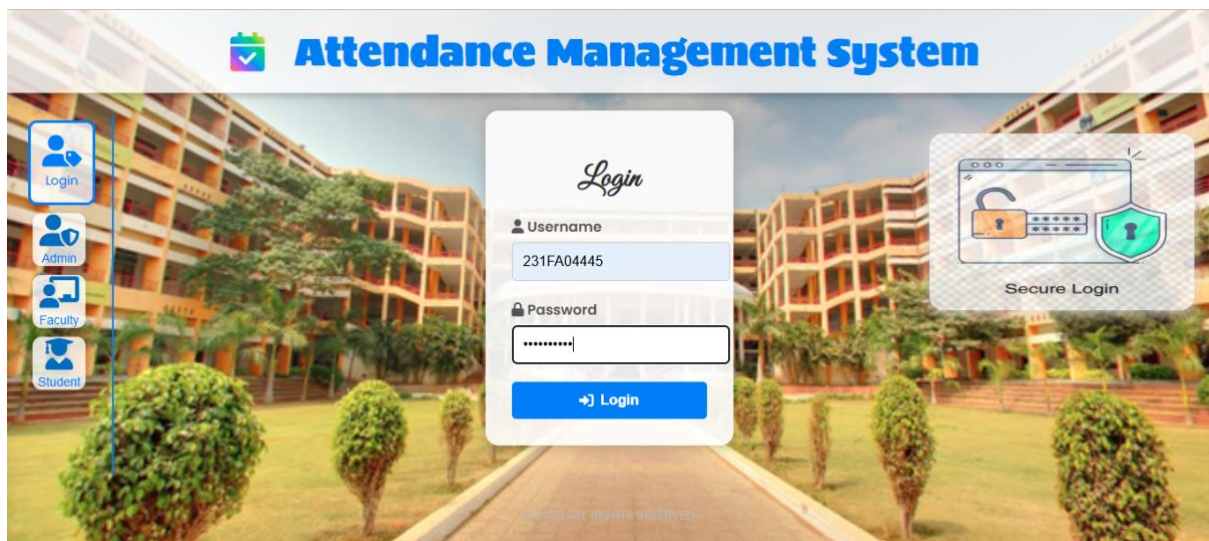


Faculty login page:



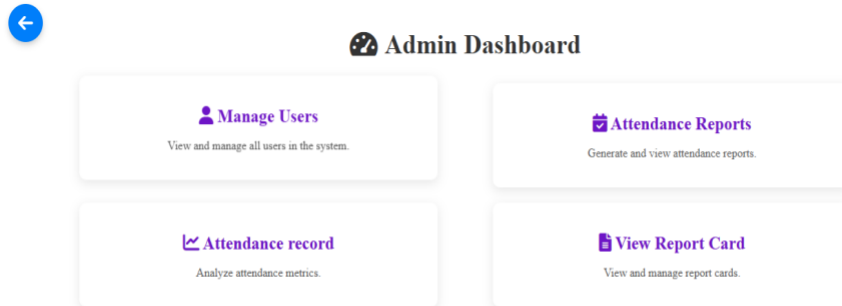
The image shows the Faculty login page for the Attendance Management System. The page features a background image of a school building. On the left, there is a vertical sidebar with icons for Login, Admin, Faculty, and Student. The main content area has a central login form titled "Login" with a cursive font. The form includes fields for Username (containing "022") and Password (containing masked characters "****"). A blue "Login" button is at the bottom of the form. To the right of the form, there is a graphic titled "ATTENDANCE MANAGEMENT SYSTEM" showing a calendar and a list of students with checkmarks. The top of the page has a header with a calendar icon and the text "Attendance Management System".

Student login page:

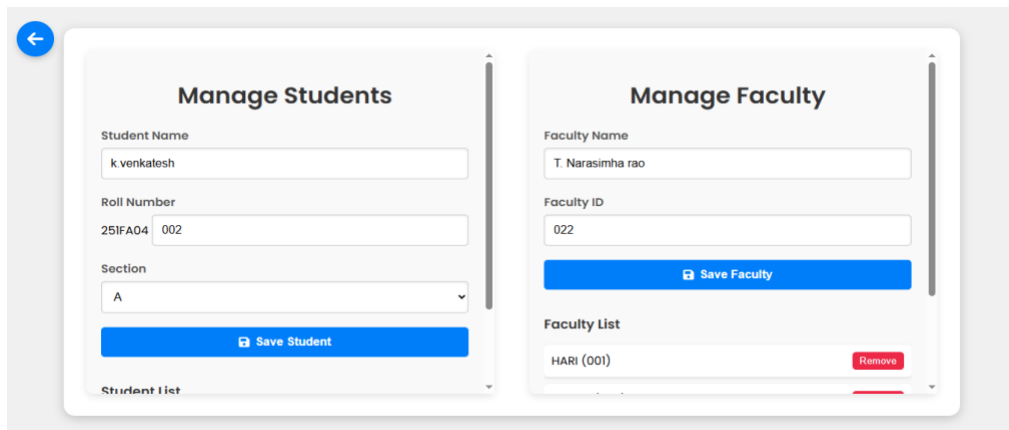


The image shows the Student login page for the Attendance Management System. The page features a background image of a school building. On the left, there is a vertical sidebar with icons for Login, Admin, Faculty, and Student. The main content area has a central login form titled "Login" with a cursive font. The form includes fields for Username (containing "231FA04445") and Password (containing masked characters "*****"). A blue "Login" button is at the bottom of the form. To the right of the form, there is a graphic titled "Secure Login" showing a shield and a key. The top of the page has a header with a calendar icon and the text "Attendance Management System".

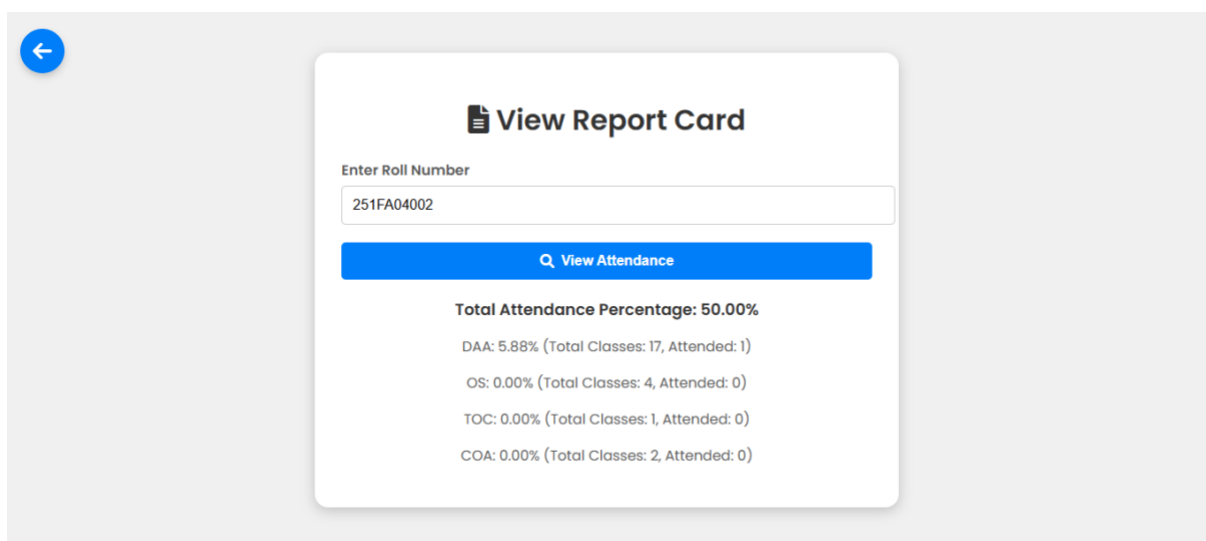
Admin Dashboard:



Manage Users:



View report card:



Attendance report:

←

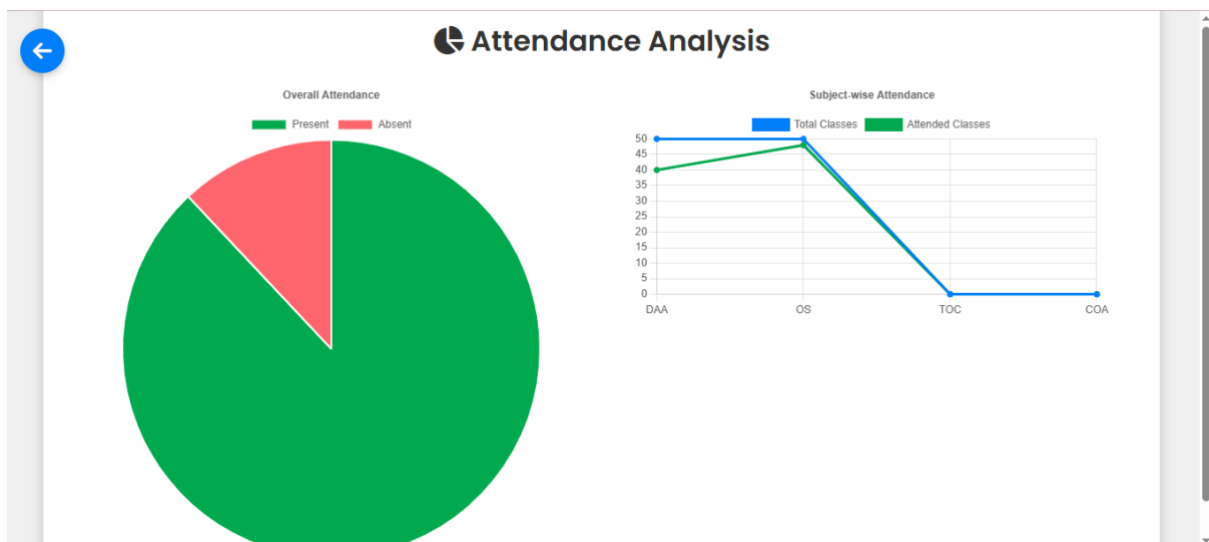
Attendance Details

Clear Attendance Data

Select Section: A

Roll Number	Period	Subject	Date	Status
25IFA04001	1	DAA	2025-03-17	Present
25IFA04002	1	DAA	2025-03-17	Present
25IFA04003	1	DAA	2025-03-17	Present
25IFA04004	1	DAA	2025-03-17	Present
25IFA04005	1	DAA	2025-03-17	Present
25IFA04006	1	DAA	2025-03-17	Present
25IFA04007	1	DAA	2025-03-17	Present

Attendance record:



6.Conclusion:

The **Attendance Management System (AMS)** successfully addresses the need for efficient and accurate tracking of attendance and leave records. It provides a user-friendly interface for both administrators and users, ensuring seamless registration, login, and attendance marking. The system offers strong security through protecting sensitive data and preventing unauthorized access. Additionally, it ensures accurate attendance records, handles edge cases effectively, and generates reports for administrators. With scalable functionality, the system can support organizations of various sizes. While it meets the current requirements, future enhancements such as mobile integration, automated notifications, and advanced analytics could further improve its capabilities. Overall, the AMS is a reliable solution for managing attendance efficiently while offering room for future improvements.

References:

- 1.<https://www.semanticscholar.org/paper/Attendance-management-system-Shailendra-Singh/f077c4af9081f2d7468c4288457e9fa726e4b282>
- 2.https://www.researchgate.net/publication/380239538_Attendance_Management_System_Project
- 3.<https://www.semanticscholar.org/paper/Attendance-management-system-Shailendra-Singh/f077c4af9081f2d7468c4288457e9fa726e4b282>

Project Link: <https://vuams.netlify.app/>